

FAST — National University of Computer and Emerging Sciences

Karachi Campus

Department of Software Engineering

PF LAB TASKS

Course Code:	C1002
Subject:	Programming Fundamentals
Section:	D & C
Instructor:	Mr. Shayan Ali

STUDENT SUBMISSION DETAILS

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Roll No / ID:	26K-3131
Class / Section: D	BSSE-1D
Date of Submission:	24-09-2026

Submission Guidelines

1. Write and compile each program separately in a C compiler (e.g. Code::Blocks, Dev-C++, or an online IDE).
2. Test each program with more than one input where relevant, to confirm all conditions work correctly.
3. Submit all tasks before the given deadline.

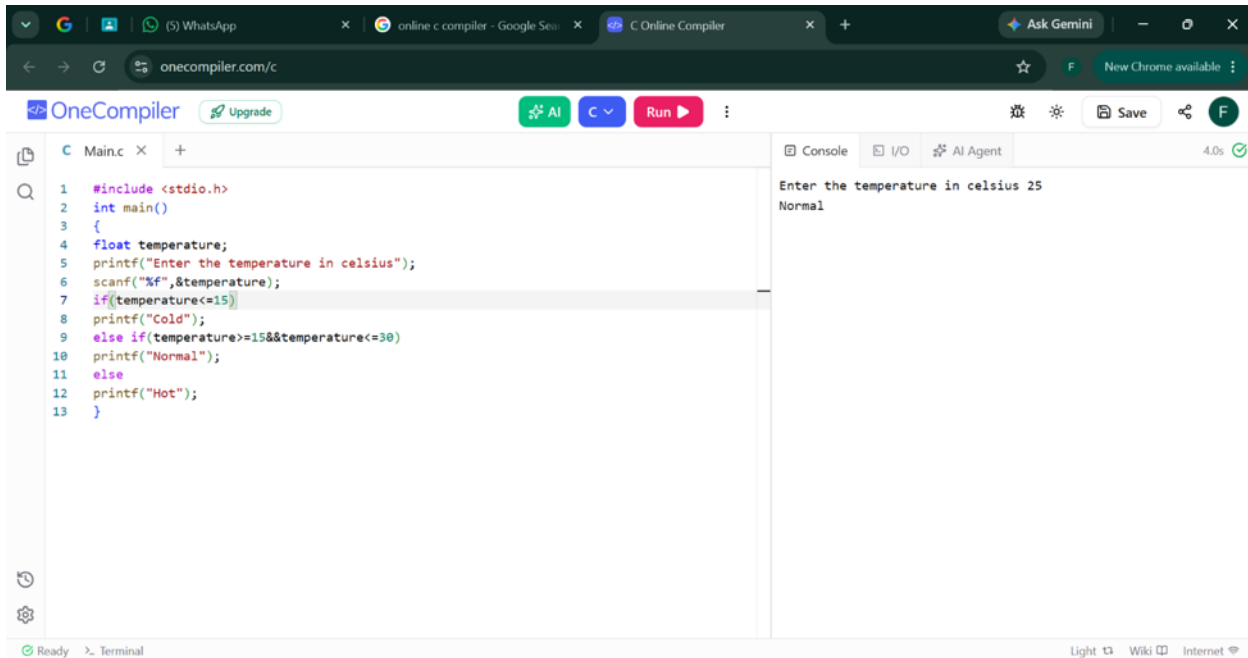
TASK # 01 — Temperature Status *(if-else)*

A weather monitoring system takes the temperature in Celsius from the user. Write a C program using if-else to display "Cold" if the temperature is below 15, "Normal" if the temperature is between 15 and 30, and "Hot" if the temperature is above 30.

```
#include <stdio.h>

int main()
{
    float temperature;
    printf("Enter the temperature in celsius");
    scanf("%f",&temperature);
    if(temperature<=15)
        printf("Cold");
    else if(temperature>=15&&temperature<=30)
```

```
printf("Normal");  
  
else  
  
printf("Hot");  
  
}
```

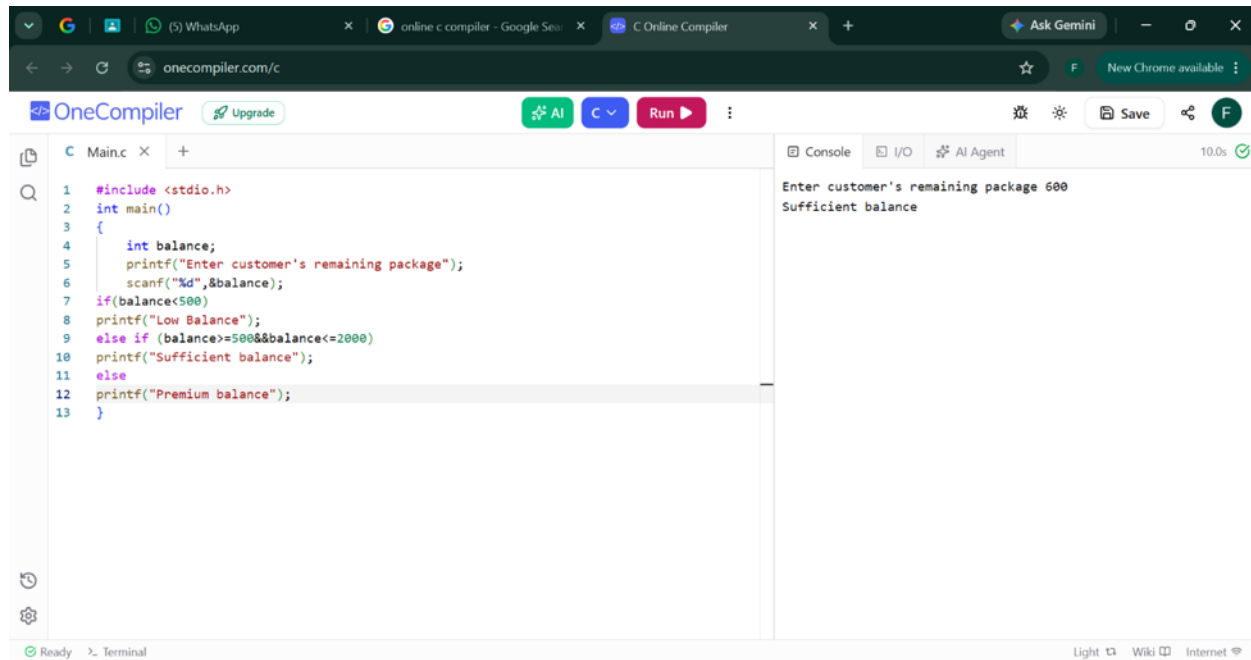


TASK # 02 — Mobile Data Package (if-else)

A mobile service provider checks a customer's remaining balance before activating a package. Write a C program that takes the balance as an integer and uses if-else to display "Low Balance" if the balance is below 500, "Sufficient Balance" if it is between 500 and 2000, and "Premium Balance" if it is above 2000.

```
#include <stdio.h>  
  
int main()  
{  
    int balance;  
  
    printf("Enter customer's remaining package");  
  
    scanf("%d",&balance);  
  
    if(balance<500)  
        printf("Low Balance");  
  
    else if (balance>=500&&balance<=2000)  
        printf("Sufficient balance");  
  
    else
```

```
printf("Premium balance");
}
```



TASK # 03 — Hospital Appointment (*nested if-else*)

A hospital appointment system asks the patient to enter three integer values: appointment, doctorAvailable, and registrationCompleted, where 1 means yes and 0 means no. Write a C program using nested if-else statements to first check whether the patient has an appointment, then check whether the doctor is available, and finally check whether the registration is completed before displaying whether the patient can meet the doctor.

```
#include <stdio.h>

int main()
{ int Appointment,Doctor_available,Registration_complete;

printf("Enter Appoinment,Doctor_avaiable,Registration_complete(0 for No,1 for Yes): ");
scanf("%d %d %d",&Appointment, &Doctor_available, &Registration_complete);

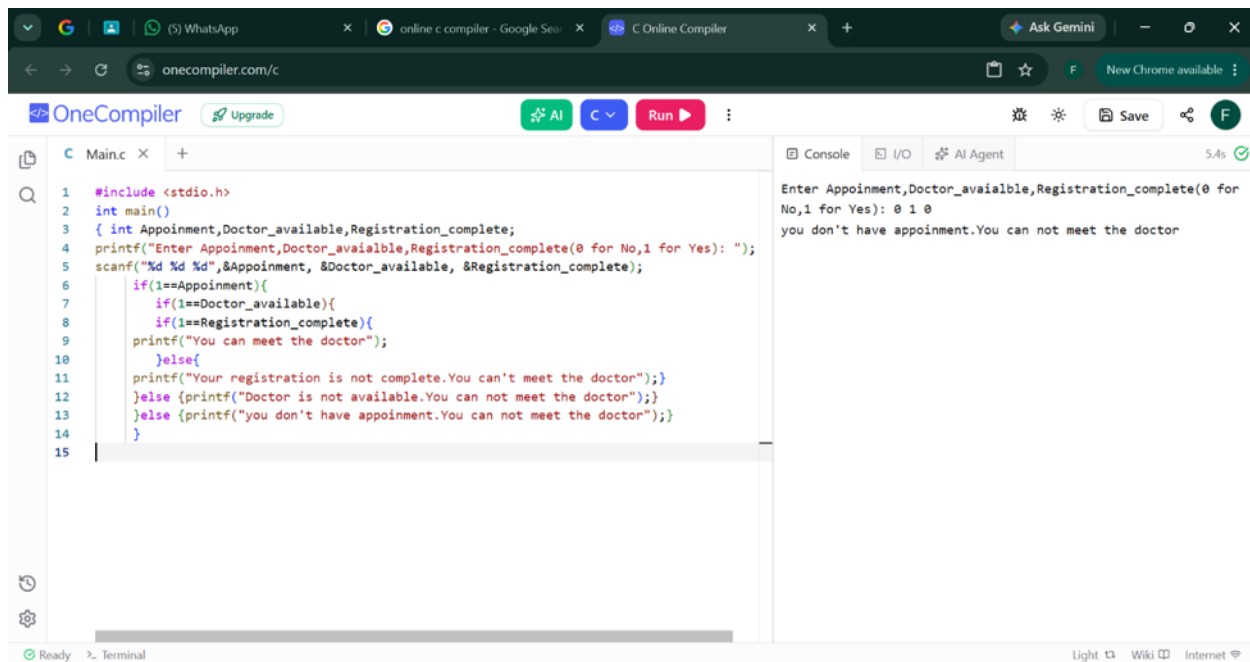
    if(1==Appointment){
        if(1==Doctor_available){
            if(1==Registration_complete){
                printf("You can meet the doctor");
            }else{
                printf("Your registration is not complete.You can't meet the doctor");}
        }else {printf("Doctor is not available.You can not meet the doctor");}
```

```

    }else {printf("you don't have appointment.You can not meet the doctor");}

}

```



TASK # 04 — Online Food Order (*nested if-else*)

An online food delivery system asks the customer to enter three integer values: restaurantOpen, itemAvailable, and balanceSufficient, where 1 means yes and 0 means no. Write a C program using nested if-else statements to first check whether the restaurant is open, then whether the selected item is available, and finally whether the customer has sufficient balance before displaying the appropriate order status.

```
#include <stdio.h>
```

```
int main() {
```

```
    int restaurant_open, item_available, balance_sufficient;
```

```
    printf("Enter restaurant status, item availability and balance status (0 for No, 1 for Yes): ");
```

```
    scanf("%d %d %d", &restaurant_open, &item_available, &balance_sufficient);
```

```
    if (restaurant_open == 1) {
```

```
        if (item_available == 1) {
```

```
            if (balance_sufficient == 1) {
```

```
                printf("Your order has been placed");
```

```
            } else {
```

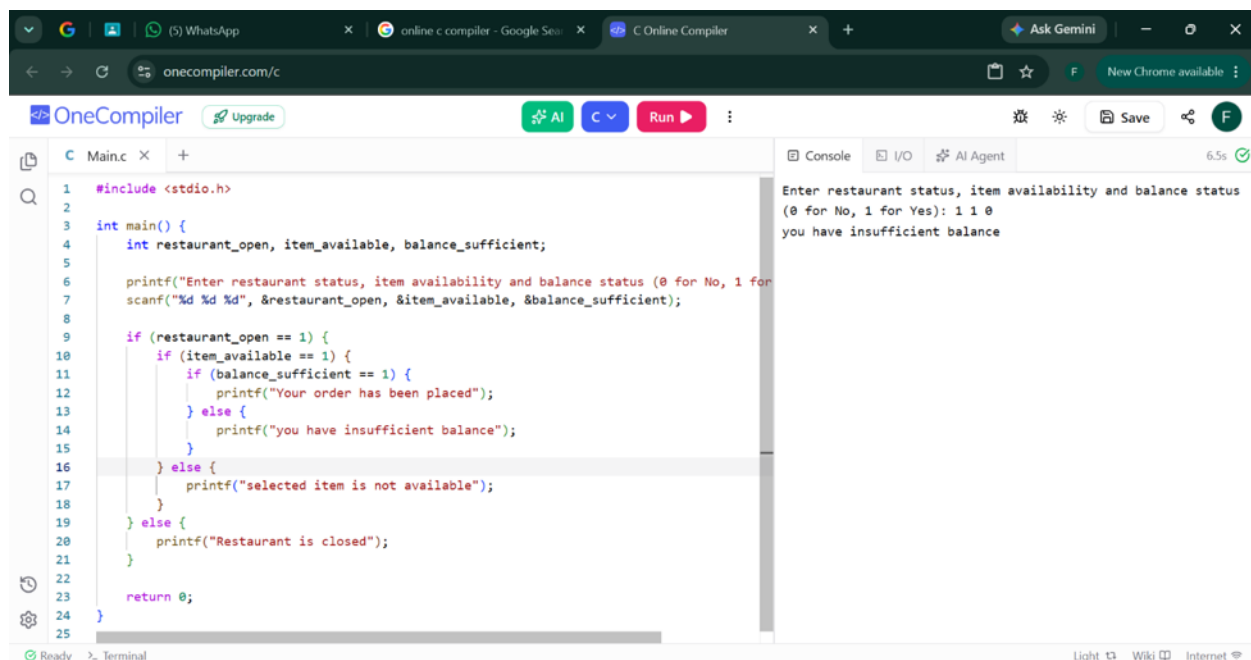
```
                printf("you have insufficient balance");
```

```

    }
} else {
    printf("selected item is not available");
}
} else {
    printf("Restaurant is closed");
}

return 0;
}

```



TASK # 05 — ATM and Transaction Selection (Nested switch):

An ATM provides four operations: 1 for Balance Inquiry, 2 for Cash Withdrawal, 3 for Cash Deposit, and 4 for PIN Change. After the user selects an operation, the ATM displays a second menu related to that operation; for example, after selecting Cash Withdrawal, the user can choose 1 for Savings Account or 2 for Current Account. Write a C program using a **nested switch statement** to take both choices as integer inputs and display the selected operation and account type

```
#include <stdio.h>
```

```

int main() {
    int mainChoice, subChoice;

    printf("ATM Menu:\n");

    printf("1. Balance Inquiry\n");
    printf("2. Cash Withdrawal\n");

```

```
printf("3. Cash Deposit\n");
printf("4. PIN Change\n");
printf("Enter your choice (1-4): ");
scanf("%d", &mainChoice);
switch (mainChoice) {
    case 1:
        printf("You selected: Balance Inquiry\n");
        break;
    case 2:
        printf("You selected: Cash Withdrawal\n");
        printf("Select Account Type:\n");
        printf("1. Savings Account\n");
        printf("2. Current Account\n");
        printf("Enter your account choice (1-2): ");
        scanf("%d", &subChoice);
        switch (subChoice) {
            case 1:
                printf("Account Type: Savings Account\n");
                break;
            case 2:
                printf("Account Type: Current Account\n");
                break;
            default:
                printf("Invalid Account Type!\n");
                break;
        }
        break;
    case 3:
        printf("You selected: Cash Deposit\n");
        break;
    case 4:
        printf("You selected: PIN Change\n");
        break;
    default:
```

```

        printf("Invalid Operation Choice!\n");

        break;

    }

    return 0;

}

```

The screenshot shows the OneCompiler C IDE with the following code in `Main.c`:

```

1  #include <stdio.h>
2  int main() {
3      int mainChoice, subChoice;
4      printf("ATM Menu:\n");
5      printf("1. Balance Inquiry\n");
6      printf("2. Cash Withdrawal\n");
7      printf("3. Cash Deposit\n");
8      printf("4. PIN Change\n");
9      printf("Enter your choice (1-4): ");
10     scanf("%d", &mainChoice);
11     switch (mainChoice) {
12         case 1:
13             printf("You selected: Balance Inquiry\n");
14             break;
15         case 2:
16             printf("You selected: Cash Withdrawal\n");
17             printf("Select Account Type:\n");
18             printf("1. Savings Account\n");
19             printf("2. Current Account\n");
20             printf("Enter your account choice (1-2): ");
21             scanf("%d", &subChoice);
22             switch (subChoice) {
23                 case 1:
24                     printf("Account Type: Savings Account\n");
25                     break;

```

The console output on the right shows the execution results:

```

ATM Menu:
1. Balance Inquiry
2. Cash Withdrawal
3. Cash Deposit
4. PIN Change
Enter your choice (1-4): 2
You selected: Cash Withdrawal
Select Account Type:
1. Savings Account
2. Current Account
Enter your account choice (1-2): 1
Account Type: Savings Account

```

The screenshot shows the OneCompiler C IDE with the following code in `Main.c`:

```

2  int main() {
11     switch (mainChoice) {
15         case 2:
22             switch (subChoice) {
23                 case 1:
24                     printf("Account Type: Savings Account\n");
25                     break;
26                 case 2:
27                     printf("Account Type: Current Account\n");
28                     break;
29                 default:
30                     printf("Invalid Account Type!\n");
31                     break;
32             }
33             break;
34         case 3:
35             printf("You selected: Cash Deposit\n");
36             break;
37         case 4:
38             printf("You selected: PIN Change\n");
39             break;
40         default:
41             printf("Invalid Operation Choice!\n");
42             break;
43     }

```

The console output on the right shows the execution results:

```

ATM Menu:
1. Balance Inquiry
2. Cash Withdrawal
3. Cash Deposit
4. PIN Change
Enter your choice (1-4): 2
You selected: Cash Withdrawal
Select Account Type:
1. Savings Account
2. Current Account
Enter your account choice (1-2): 1
Account Type: Savings Account

```